

Project Continuity Summary: Loan Approval Prediction

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Batch: D

1. Business Problem & Objective

The core business problem is to automate and optimize the loan approval process for financial institutions. Manual underwriting is time-consuming and prone to risk. The objective of this project is to analyse historical loan application data to identify the key drivers of loan approval and prepare a robust dataset for predictive machine learning, ultimately reducing default rates and improving operational efficiency.

2. Target Variable

The target variable is Loan_Status (1 = Approved, 0 = Rejected). This frames the project as a binary classification problem.

3. Review of Previous Work (Weeks 1 & 2)

Initial Findings: Early inspection revealed significant data quality issues, including missing values in critical fields like Credit_History and LoanAmount, severe right-skewness in income variables, and extreme outliers.

Data Quality & Preprocessing (Week 2):

Missing Values: Imputed using the Mode for categorical variables and the Median for skewed numerical variables.

Outliers: Capped using the IQR (Interquartile Range) method to retain data points while limiting extreme variance.

Encoding: Applied Binary Mapping for Yes/No variables and One-Hot Encoding for Property_Area.

Initial Feature Engineering: Created TotalIncome, Loan_to_Income_Ratio, and EMI to provide stronger financial signals.

Scaling: Applied StandardScaler to prepare the final ML-ready dataset.

4. Remaining Questions for Week 3 (Advanced Analysis)

While Week 2 successfully cleaned the data, several critical questions remain that require statistical validation before modelling:

Is the difference in TotalIncome and LoanAmount between approved and rejected loans statistically significant, or due to random chance?

How strongly does Credit_History correlate with Loan_Status in a statistically proven manner?

Do the newly engineered features (Loan_to_Income_Ratio, EMI) actually provide better predictive power than the original raw income features?

Are there hidden multivariate relationships (e.g., between Property Area, Income, and Approval) that basic EDA missed?

Week 3 will address these questions through advanced EDA, rigorous hypothesis testing, and further feature refinement to finalize the dataset for Week 4 (Machine Learning).